Predicting Formal Protest Petitions Against Housing Development

Evidence from Austin, Texas

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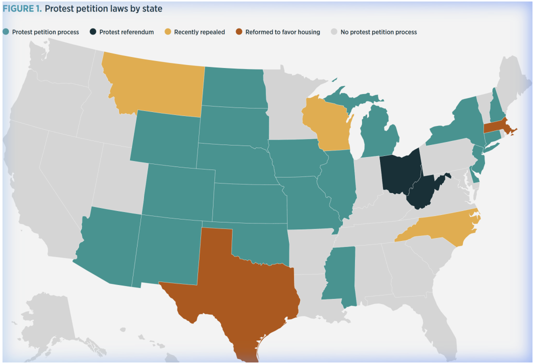
Institutionalized NIMBYism

The 20% Threshold

If owners of 20% of the land within 200 feet of a proposed rezoning file written opposition, City Council requires a **supermajority (3/4)** vote to approve it.

This effectively gives small groups of nearby owners veto power over citywide housing decisions.



Statutory Context: The Institutional Veto

Texas LGC Chapter 211

- **The 20% Rule**: If owners of 20% of the land within 200ft sign a valid protest, a *supermajority* vote is required.
- **High-Stakes Voting**: In a 7-member council, this moves from 4 votes to 6.
- **Asymmetric Veto**: A localized minority can block citywide housing policy through formal administrative friction.

PETITION

Case Number: **C14-2007-0131** Date: July 29, 2008

1506 WALLER ST, 807 E 16TH ST & 908 E 15TH ST

Total Area Within 200' of Subject Tract 293,002.55

1	02-0906-0502	MONAHAN CASEY J	7423.26	2.53%
2	02-0906-0503	MONAHAN CASEY J	7390.73	2.52%
3	02-0906-0504	CLARK MICHAEL G & RITA S DE BE	7354.85	2.51%
4	02-0906-0506	MARTINSON KATHY L	1645.43	0.56%
5	02-0906-0507	CHILES ROSALIE BENSON	1618.17	0.55%
6	02-0906-0508	KOCHERT KELLEY	1612.79	0.55%
7	02-0906-0509	MONAHAN CASEY J	1419.72	0.48%
8	02-0906-0606	TIMMERMANN TERRELL	3452.33	1.18%
9	02-0906-0607	MERCADO FREDDY G & MARIA R	9482.82	3.24%
10	02-0906-0901	LEGETT GEORGIA F	4459.36	1.52%
11	02-0906-0911	LEGETT GEORGIA F	7856.05	2.68%
12	02-0906-1001	MEDINA JAMES & KRISTINE M GARA	13204.88	4.51%
13	02-0906-1011	RECER DANALYNN	9454.11	3.23%
14	02-0906-1013	BRINSMADE LOUISA C	7634.37	2.61%
15				0.00%
16				0.00%
17				0.00%
18				0.00%
19				0.00%
20				0.00%
21				0.00%
22				0.00%
23				0.00%
24				0.00%
25				0.00%
26				0.00%
27				0.00%
Validated By:			**Total Area of Petitioner:**	**Total %**
	Stacy Meeks		**84,008.88**	**28.67%**

The Goal and The Data

The Research Question Can we anticipate formal protest petitions *before* public hearings?

Filing-Date Records

- Site area, height/FAR deltas.
- Current vs requested zoning.
- Neighborhood spatial context.

A Sample Case

Feature	Case C14-2018-0156
Gross Site Area	4.92 Acres
Height Request	+15.5 ft
FAR Request	+0.43
Neighborhood	Riverside
Result	**Valid Petition**

Technical Pipeline: Data Generation

As-Of Context Reconstruction

- **Data Recovery**: Reconstructing 15+ years of zoning agendas.
- **Parsing Pipelines**: PDF scraping of Clerk records.
- **Timeline Integrity**: Ensuring no “future leakage” (predicting only with data available at filing).
- **Expert Grounding**: Semi-structured interviews ($N = 8$) with planners and advocates to calibrate model interpretations.

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Research Project Overview

Predicting NIMBYism in Austin's Housing Reform Era

Project Abstract

This research forecasts opposition to development using AI and interprets the mechanics of that opposition in Austin, Texas. By analyzing petition protest letters, public hearing transcripts, and zoning application data, the study evaluates the democratic implications of using predictive analytics to anticipate political behavior. The project employs predictive modeling, natural language processing, and qualitative analysis to predict opposition behavior, identify recurring patterns in opposition rhetoric, and examine their association with planning outcomes.

Study Objectives

Identify Patterns: Characterize the opposition rhetoric and arguments used in protest petitions to oppose rezoning and identify demographic predictors of this opposition.

Evaluate Predictive Tools: Assess whether administrative data and predictive models can accurately forecast opposition to specific development projects.

Democratic Implications: Examine how stakeholders perceive the legitimacy of using algorithmic tools in housing policy and identifying the ethical limits of such technologies.

Participant Role

We are seeking perspectives from urban planners, developers, neighborhood association leaders, and community organizers.

Activity: A 45-60 minute one-on-one interview conducted virtually or in person.

Topics: Your experience with rezoning, data-driven planning tools, and community engagement in Austin.

Data Privacy and Ethical Standards

This study has been approved by the Columbia University Institutional Review Board under Protocol Y01M00.

Participation is entirely voluntary.

You may decline to answer any specific question or discontinue the interview at any time.

Identities will be kept strictly confidential in resulting publications unless you grant explicit permission to be named.

Researcher Contact **IRB Contact**

The Rarity Problem

Sample Constraints

- **Universe**: 2007–2024 Austin rezonings.
- **Distribution**: $\approx 2.6\%$ baseline petition rate.
- **Evaluation**: Strictly temporal (forward-looking) holdouts.

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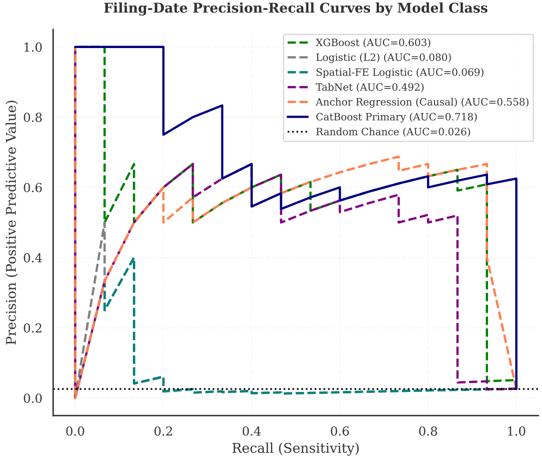
Predicting Resistance

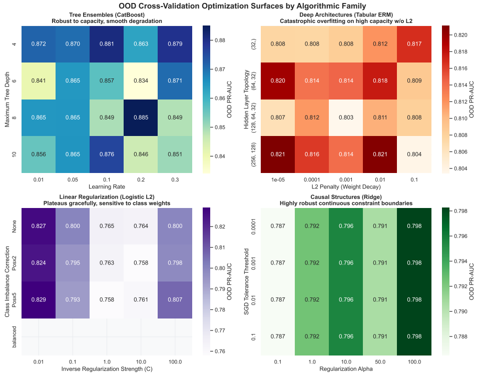
Trees Dominate at the Top of the Ranking



Modeling Architectures: Benchmarking

CatBoost vs. TabNet

- **Tree Ensembles**: Robust to diverse feature scales; dominant in tabular planning data.
- **Deep Architectures**: High sensitivity to parameter tuning; struggle with discrete spatial boundaries.
- **The Winner**: Tree-based models achieve 2X the precision in high-triage tiers.

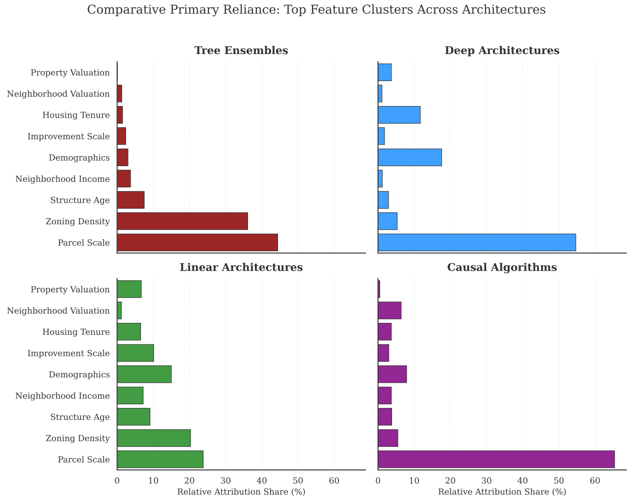


Dealing with Collinearity

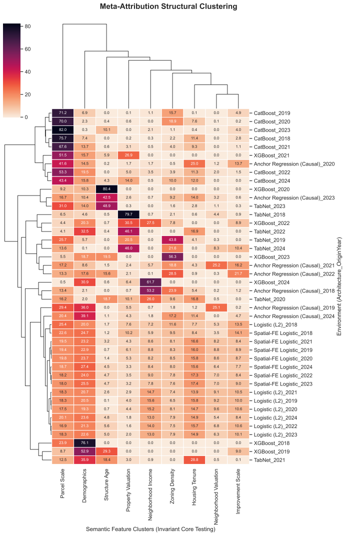
Semantic Clustering

Standard feature attribution fragments importance across collinear features because isolated real estate variables proxy identical physical concepts (e.g., site area, deed acreage).

We sum absolute TreeSHAP values within semantic clusters of correlated features.



Architectural Separation of Feature Reliance

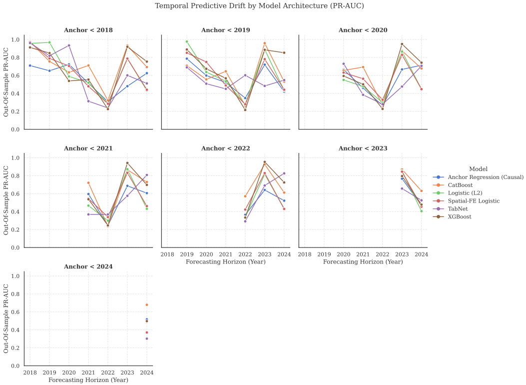


Temporal Drift

When does the model break?

Historical models suffer from *concept drift* when encountering modern low-conflict environments.

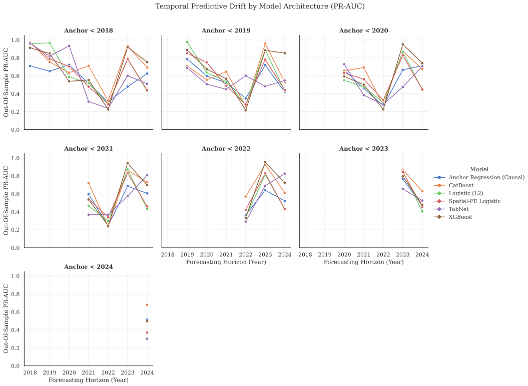
Linear models collapse entirely as the gap between training and test years widens, while Tree Ensembles degrade more gracefully.



Agency in Temporal Shifts

The 2022 Performance Dip

- **The Finding**: Model performance plummeted in 2022 across all architectures.
- **Hypothesis**: Consistently low performance in 2022 align with pandemic-era filing shifts and administrative uncertainty.
- **Agency Proof**: Proves that predictability is not static, but contingent on social/political intelligence.



Qualitative Context: Stakeholder Insights

The “Motive Blindness” Discovery

- The algorithm cannot distinguish between **exclusionary NIMBYism** and **anti-displacement resistance**.
- Quantitative risk weights are independent of neighborhood intent or social equity goals.

Expert Perspectives

- *“Haters are overrepresented and silent supporters are underrepresented”* (Beaudet).
- Algorithm fails *“to identify where people that are in actual need actually are”* (Torrado).
- Professionals view rigid probability cutoffs as “ethically hazardous.”

COLUMBIA UNIVERSITY
Graduate School of Architecture, Planning and Preservation — Semi-Structured Stakeholder Interview Guide

Predicting NIMBYism in Austin’s Housing Reform Era

Purpose

To explore how key stakeholders understand neighborhood opposition to housing development in Austin and how they view the potential use of predictive or algorithmic tools in planning and zoning.

Opening

1. Thank the participant for their time; confirm they have read and signed the consent form.
2. Confirm whether they consent to audio recording.
3. Remind them they may skip any question or stop at any time.

Part 1: Background and Role

1. Can you briefly describe your current role and how it relates to housing, planning, or land use in Austin?
2. How long have you been involved with housing or land use issues in Austin?
3. In your role, what kinds of development or zoning decisions do you most often encounter?

Part 2: Experiences with Neighborhood Opposition

1. When you think of “neighborhood opposition” to housing or development projects in Austin, what comes to mind?
2. Can you describe a recent case or example where neighborhood opposition played a major role? What happened?
3. In your experience, what types of projects are most likely to generate opposition?
4. Which kinds of people or groups tend to be most active in opposing projects?

Part 3: Perceptions of Fairness and Participation

1. Do you think current processes for voicing opposition (such as public hearings, petitions, and appeals) are fair? Why or why not?
2. Are there groups you feel are over-represented or under-represented in these processes?
3. How do you think opposition affects whose interests are ultimately reflected in housing and zoning decisions?

Part 4: Views on Predictive and Algorithmic Tools

1. Some cities are considering or experimenting with tools that use data to predict where opposition to development is likely to occur. Have you encountered ideas like this before?
2. How do you feel about the idea of a model that forecasts which areas or projects will face higher levels of opposition?
3. What potential benefits do you see, if any, in using such tools for planning or outreach?
4. What potential risks or harms concern you most? For example, concerns about profiling, surveillance, or chilling participation.
5. Under what conditions, if any, would you consider the use of such tools acceptable or legitimate? (e.g., transparency, public input, limits on use).

Conclusion

What we showed

- **Predictive Utility**: Filing-date administrative data carries real predictive signal (PR-AUC 0.724), **qualitatively validated by practitioners**.
- **Tree Dominance**: Ensemble methods consistently lead at the top of the risk ranking tier.

What surprised us

- **Observed Ceilings**: Model reliability is bounded by something other than longitudinal feature richness.
- **Defensible Agency**: Performance floors are consistent with mobilization being a **strategic choice** rather than a mechanical response to parcel spatiality.

What it implies

- **Decision Policy**: Algorithm selection is a non-technical choice about which urban feature clusters are prioritized.

Ethics Warning: This model ranks risk; it should not be used to pre-screen sites for developers seeking the path of least resistance.

Implications: A Now-Closed Policy Window

The HB 24 Shift (2025)

- **Institutional Reform**: Texas HB 24 gutted the petition mechanism for residential upzonings.
- **Context Bound**: This thesis maps a specific, historical era of minority veto power.
- **Method Portability**: The methodology travels; the parameters remain Austin-specific.

This model identifies risk in a historical architecture; current policy has moved toward broad administrative deregulation.

Discussion / Q&A

Thank you to the committee for your time and feedback.

Thank You

To all who advised, read, and commented on my thesis

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Matthew Bauer

Adam Vosburgh

Josh Begley